## SECTION A: (FLOW MEASUREMENTS)

## QUESTION 1

APRIL 2019

1.1 Below is a description of the operation of a Pitot tube.

Complete the following sentences by using the words in the list below. Write only the word(s) next to the question number (1.1.1–1.1.11) in the ANSWER BOOK.

| velocity; impact pressure; greater; obstruction; impinges; difference; impact pressure; tapping; decrease; static pressure; pressure energy |
|---|

As fluid approaches an (1.1.1 ...) in a flow stream, the (1.1.2 ...) will (1.1.3 ...) until it reaches zero at a point it (1.1.4 ...) on it .The kinetic energy is now converted to (1.1.5 ...) Thus, the pressure built up is (1.1.6 ...) than that in the free stream because of (1.1.7 ...) The difference is measured between the pressure in the tube and the (1.1.8 ...) of the stream. The static pressure is measured by a (1.1.9 ...) in the wall of the main. The (1.1.10 ...) between the pressure in the tube and the static pressure will be measured by (1.1.11 ...), and therefore by the velocity of the stream. (11)

1.2 Explain the purpose of primary flow element. (4)

1.3 Name FIVE main orifice tapping positions and explain them. (5 × 2) (10)

1.4 Name FIVE examples of differential pressure type flow rate-measuring devices. (5)

**[30]**

## QUESTION 2

Prove with the aid of the energies and the application of Bernoulli's theorem for liquids, that $Q = C.A_2.E\sqrt{2gh}$ for an orifice plate. **[15]**

## QUESTION 1: FLOW MEASUREMENT

AUGUST 2019

1.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–I) next to the question number (1.1.1–1.1.6) in the ANSWER BOOK.

| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 1.1.1 | Velocity-sensitive device that measures the volumetric flow of gases and liquids | A | primary flow element |
| | | B | square root problem |
| 1.1.2 | Instrument that measures the total volume of flow in a certain time | C | process control |
| | | D | control valve |
| 1.1.3 | Device that creates a difference in pressure in a flowing fluid | E | integrator |
| | | F | laminar flow |
| 1.1.4 | Device that controls the amount of flow of substance | G | weir |
| 1.1.5 | Rough and irregular movement of fluid | H | swirl flowmeter |
| | | I | turbulent flow |
| 1.1.6 | Obstruction placed in an open channel at a right angle to direction of flow to create a pressure head | | |

(6 x 1) (6)

1.2 Make a neat, labelled sketch of a drag-body flowmeter. (7)

1.3 Give THREE advantages and TWO disadvantages of a drag-body flowmeter. (3 + 2) (5)

1.4 Explain, with the aid of a neat, labelled sketch, the working principle of a nutating-disc flowmeter. (4 + 7) (11)

**[29]**

## QUESTION 2: DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT

2.1 Make a neat, labelled sketch of a hand-held hydrometer. (4)

2.2 Explain the working principle of a hand-held hydrometer. (6)

2.3 Make a neat, labelled sketch of a piezoelectric hygrometer sampling system. (14 x ½) (7)

2.4 Explain the working principle of a vibrating-reed viscometer. (11)

**[28]**

## QUESTION 1

NOV 2019

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 A closed-loop controller requires an input from both the process variable and the ...

A digital variable.
B analogue variable.
C set point.
D error signal.

1.1.2 When a process is balanced, process variable ...

A is set point.
B is greater than set point.
C is less than set point
D is 50% of set point

1.1.3 Which ONE of the following pH values indicates a basic solution?

A 0
B 4
C 7
D 11

1.1.4 In oxidation reduction reactions, a substance giving up the electrons is called a/an ...

A reducing agent.
B oxidant.
C oxidising agent.
D None of the above

1.1.5 Which ONE of the following function does a primary device perform in the indirect measurement of fluid flow?

A Measures the pressure difference
B Create the pressure difference
C Measure the amount of flow
D Assures a smooth flow

(5 x 1) (5)

1.2 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–I) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 1.2.1 | A measure of water vapour present in the air | A | density |
| 1.2.2 | The temperature at which the water vapour present in the air saturates the air and begins to condense | B | saturation temperature |
| 1.2.3 | The relationship between the viscosity of gases and liquids to the steady state velocity of a known weight sphere, falling through a fluid of known density at low Reynolds number | C | non-newtonian fluids |
| 1.2.4 | When fluids are deformed by strain, the ratio between shear rate and shear stress will not be constant value | D | humidity |
| 1.2.5 | An apparatus that measures the amount of water vapour in the air | E | Newtonian fluids |
| | | F | hygrometer |
| | | G | dew point |
| | | H | hydrometer |
| | | I | Stoke's law |

(5 x 1) (5)

1.3 Complete the following sentences by using the words in the list below. Choose the answer and write only the answer next to the question number (1.3.1–1.3.3(b)) in the ANSWER BOOK

| voltage; volumetric; square root; velocity; differential pressure |
|---|

1.3.1 Turbine and magnetic flow meters are ... flow measuring devices. (1)

1.3.2 The output (a) ...of the magnetic flow meter is proportional to the (b ) ... of the fluid moving through the meter. (2)

1.3.3 All Bernoulli type primary flow elements set up a (a).... which varies as a (b) ...of the flow rate. (2)

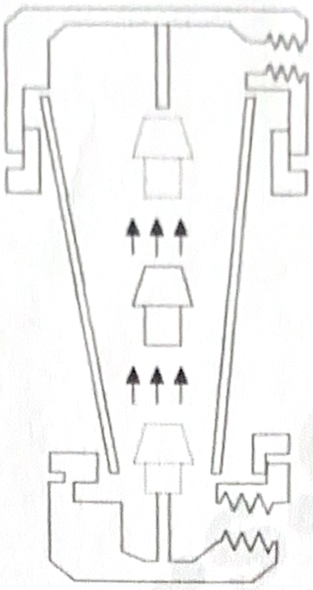
**[15]**

## SECTION A: FLOW MEASUREMENT

NOV 2020

### QUESTION 1

1.1



1.1.1 Identify the instrument above. (1)

1.1.2 Name FOUR typical applications of the instrument. (4)

1.1.3 Explain, with the aid of a sketch, how the float of the instrument above is kept clean. (4)

1.1.4 State THREE advantages and TWO disadvantages of the instrument above. (5)

1.1.5 Discuss the construction and principle of operation of the instrument above. (6)

1.2 Give FOUR disadvantages of a turbine flowmeter. (4)

**[24]**

### QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 According to Bernoulli's principle there is a relationship between fluid pressure and fluid velocity. When velocity increases, the pressure decreases and vice versa.

2.2 The eccentric orifice plate has an off-centred hole to allow undesired parts of fluid to pass through the orifice rather than build up on the upstream face.

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2.3 Vortex shedding occurs whenever a nonstreamlined object, called a bluff body, is placed in a flowing stream of gas or liquid.

2.4 The purpose of an integrator is to measure the total volume of flow in a certain time.

2.5 The purpose of secondary measuring devices is to create a difference in pressure of a flowing fluid.

(5 × 2) **[10]**

## SECTION A: FLOW MEASUREMENTS

APRIL 2021

### QUESTION 1

1.1 Suppose the flow rate of a weak acid solution is measured using a magnetic flowmeter. The conductivity of the acid is well within the acceptable range for this meter which means it works well. Now suppose the acid solution grows in strength (greater acid concentration). This will increase the conductivity of the solution, because there are now more ions available to carry an electric current. What effect will this have on the calibration of the magnetic flowmeter? Must the flowmeter be recalibrated to properly measure the acid flow again? (6)

1.2 Explain each of the FOUR energies that exist in a pipe. (4 × 2) (8)

1.3 Explain the square-root problem with reference to differential pressure and flow. (4)

1.4 What is an integral orifice differential pressure cell and where can it be used? (4)

**[22]**

### QUESTION 2

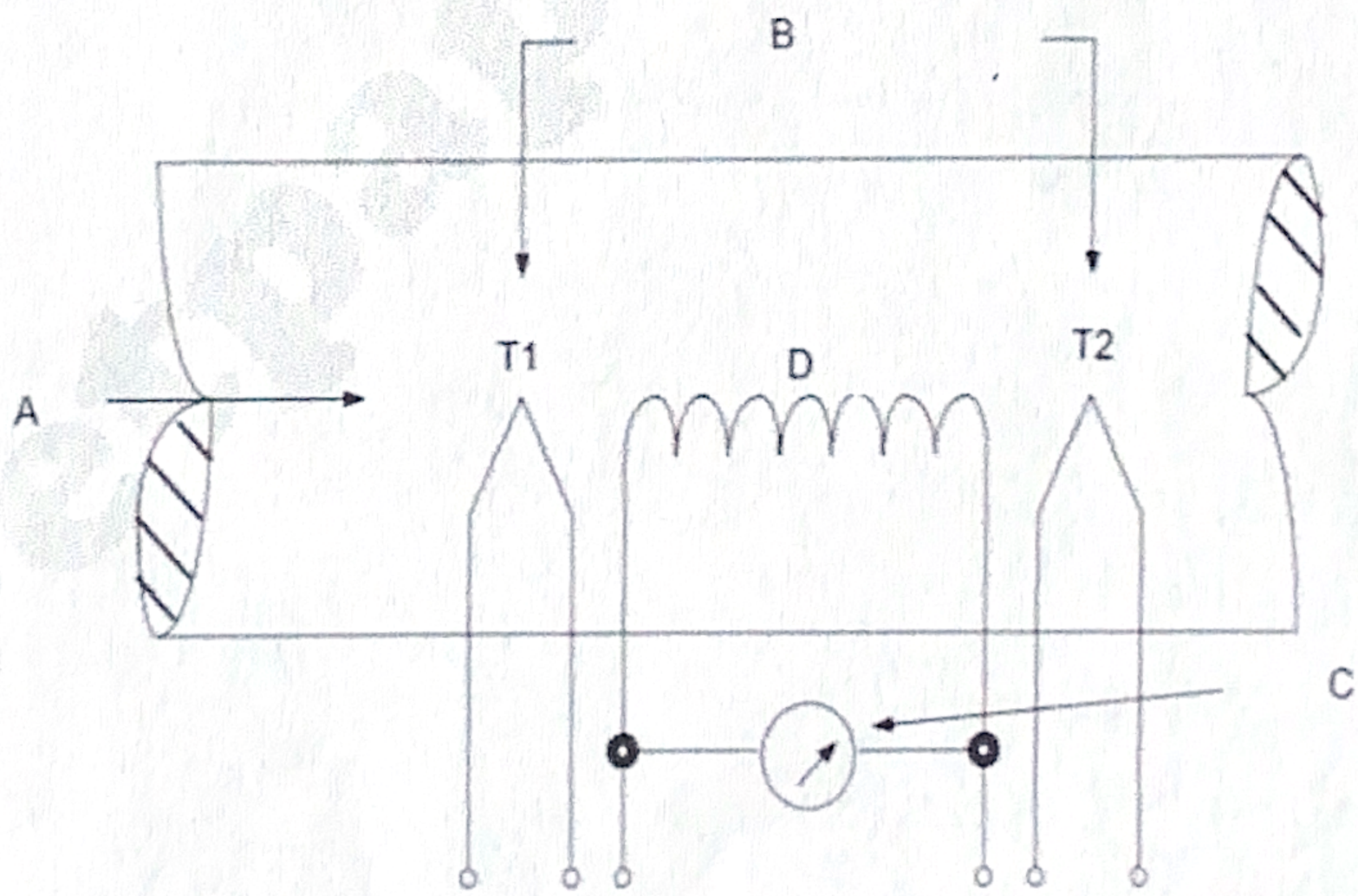



**FIGURE 1**

2.1 Label FIGURE 1 by writing the answer next to the letter (A–D) in the ANSWER BOOK. (4 × 1) (4)

2.2 Which type of flowmeter is shown in FIGURE 1? (1)

2.3 Explain the disadvantage of the flowmeter in FIGURE 1. (2)

## SECTION A: FLOW MEASUREMENT

JULY 2021

### QUESTION 1

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.1–1.7) in the ANSWER BOOK.

1.1 Bernoulli's equation is a mathematical expression of fluid density and compressibility in a restriction.

1.2 Turbine flow-measuring elements are inherently linear and require no square root extraction anywhere in the loop.

1.3 As incompressible fluid moves through a restriction, velocity increases and pressure remains the same.

1.4 For accurate operation, orifice plate flowmeters require fully developed turbulent flow.

1.5 Thermal flowmeters inherently measure maximum flow rate.

1.6 A magnetic flowmeter will properly measure the flow rate of oil.

1.7 A flag flapping in a breeze illustrates vortex shedding of dynamic fluid effects.

(7 × 1) [7]

### QUESTION 2

2.1 Name the instrument shown in FIGURE 1 below. (1)

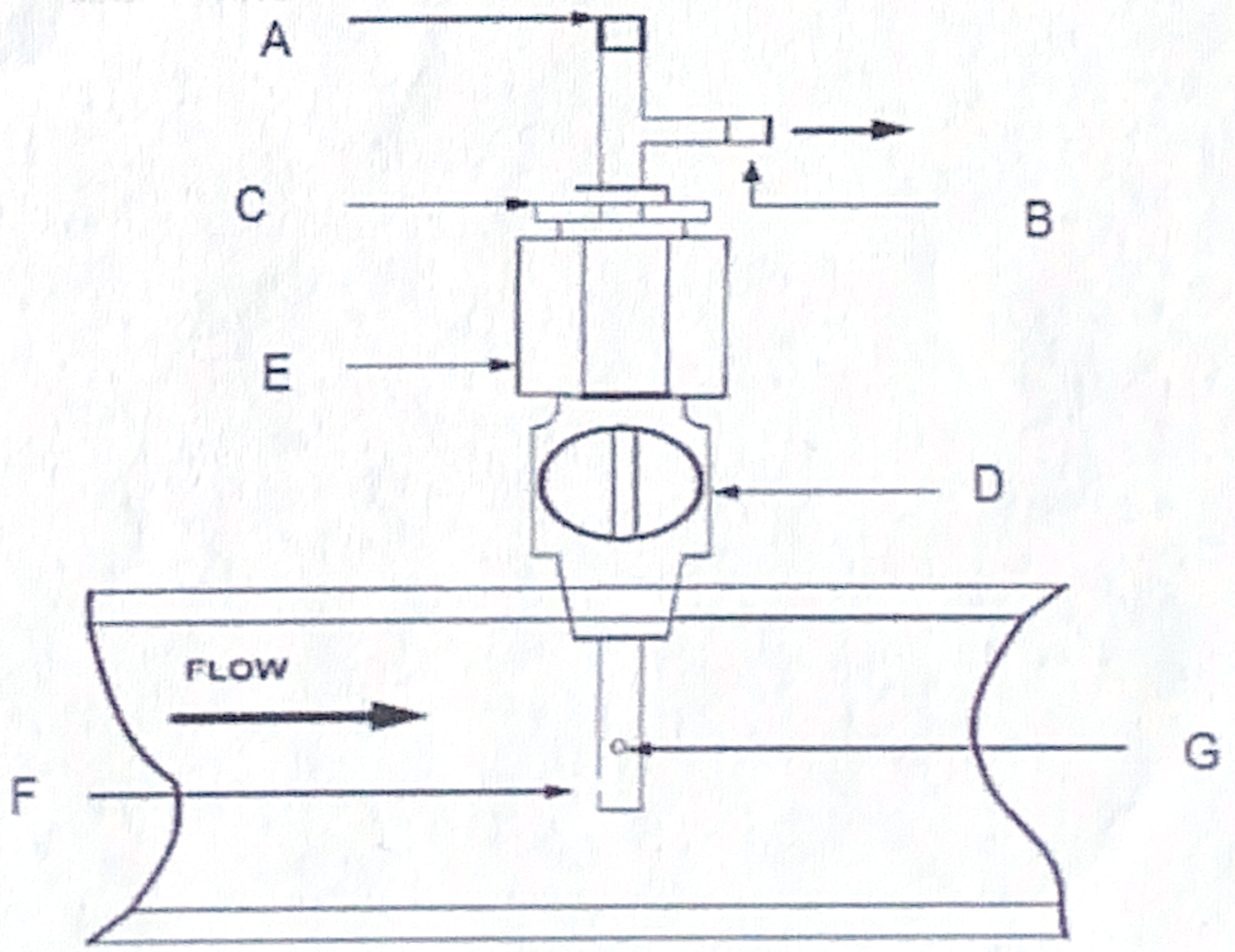



FIGURE 1

JULY 2021

2.1.1 Name components A to G. (7)

2.1.2 Explain *stagnation point*. (3)

2.2 Flowmeters are generally divided into two parts.

Name these parts. (2)

2.3 If a pipe decreases from a 9 cm diameter to 6 cm diameter and the velocity in the 9 cm section is 2,21 m/s, what is the average velocity in the 6 cm section?

**HINT:** Continuity requires that the flow rate is constant: $A_1v_1 = A_2v_2$ (7)

2.4 Weirs are used to measure flow rate primarily on open channels such as water works including irrigation, waste and sewage systems, and in pipes and conduits that are generally not completely filled with water. It is an obstruction in a flowing stream over which the liquid is forced to pass.

2.4.1 State THREE advantages and TWO disadvantages of weirs. (5)

2.4.2 Name THREE types of weirs. (3)

**[28]**

**TOTAL SECTION A: 35**

## SECTION A: FLOW MEASUREMENT

FEBRUARY 2022

### QUESTION 1

1.1 1.1.1 Which type of output is produced by a magnetic flowmeter? (2)

1.1.2 Make a neat, labelled sketch of a magnetic flowmeter. (5)

1.1.3 Name TWO liquids measured with a magnetic flowmeter. (2)

1.1.4 State Faraday's law of magnetic induction. (3)

1.2 State SEVEN factors of importance when selecting a turbine flowmeter for any given application. (7)

1.3 A venturi tube is installed on a steam line to measure the flow of high-pressure steam coming from a powerhouse boiler and going to a steam turbine to generate electricity. The contractor left you with this mess.

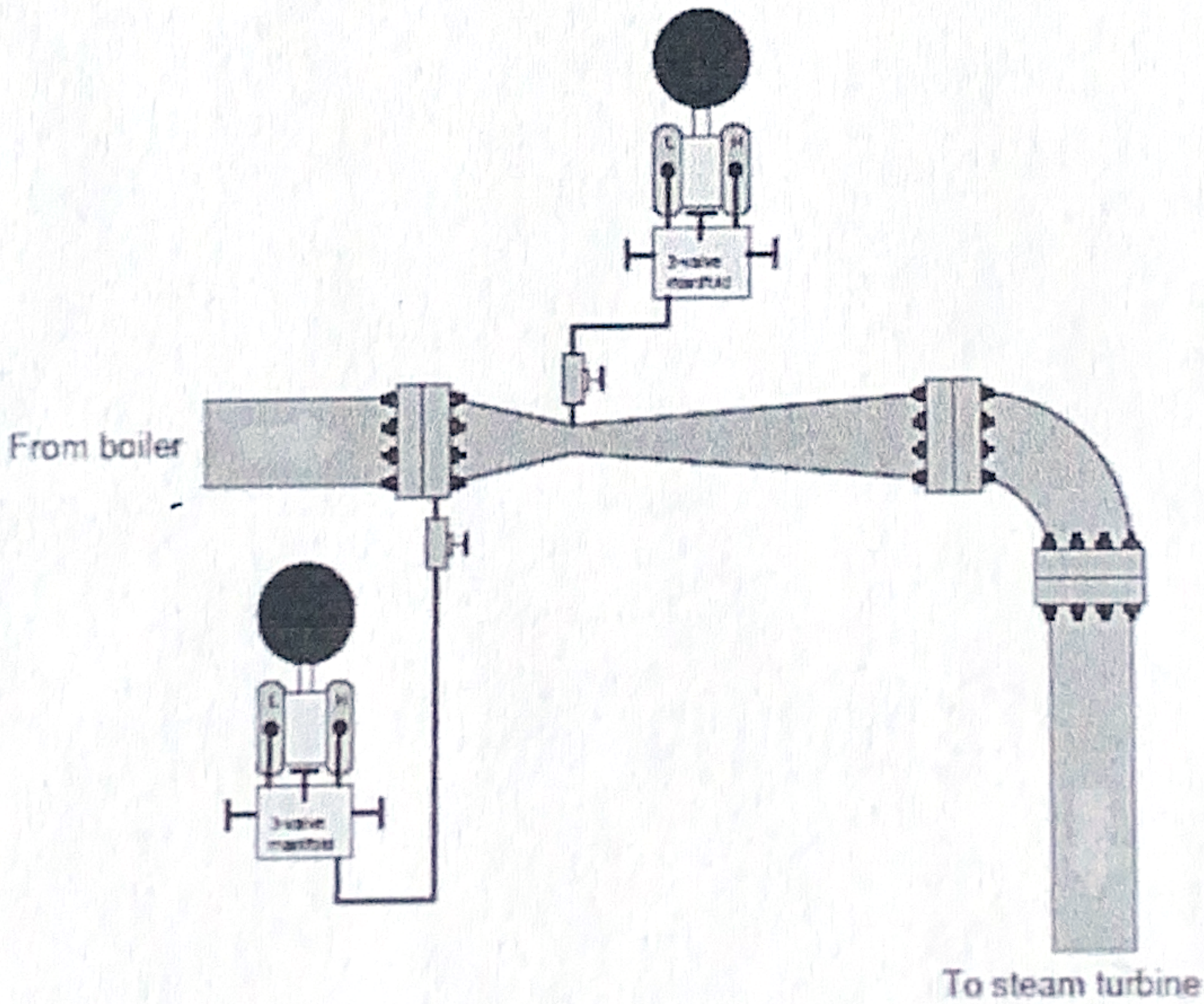



**FIGURE 1**

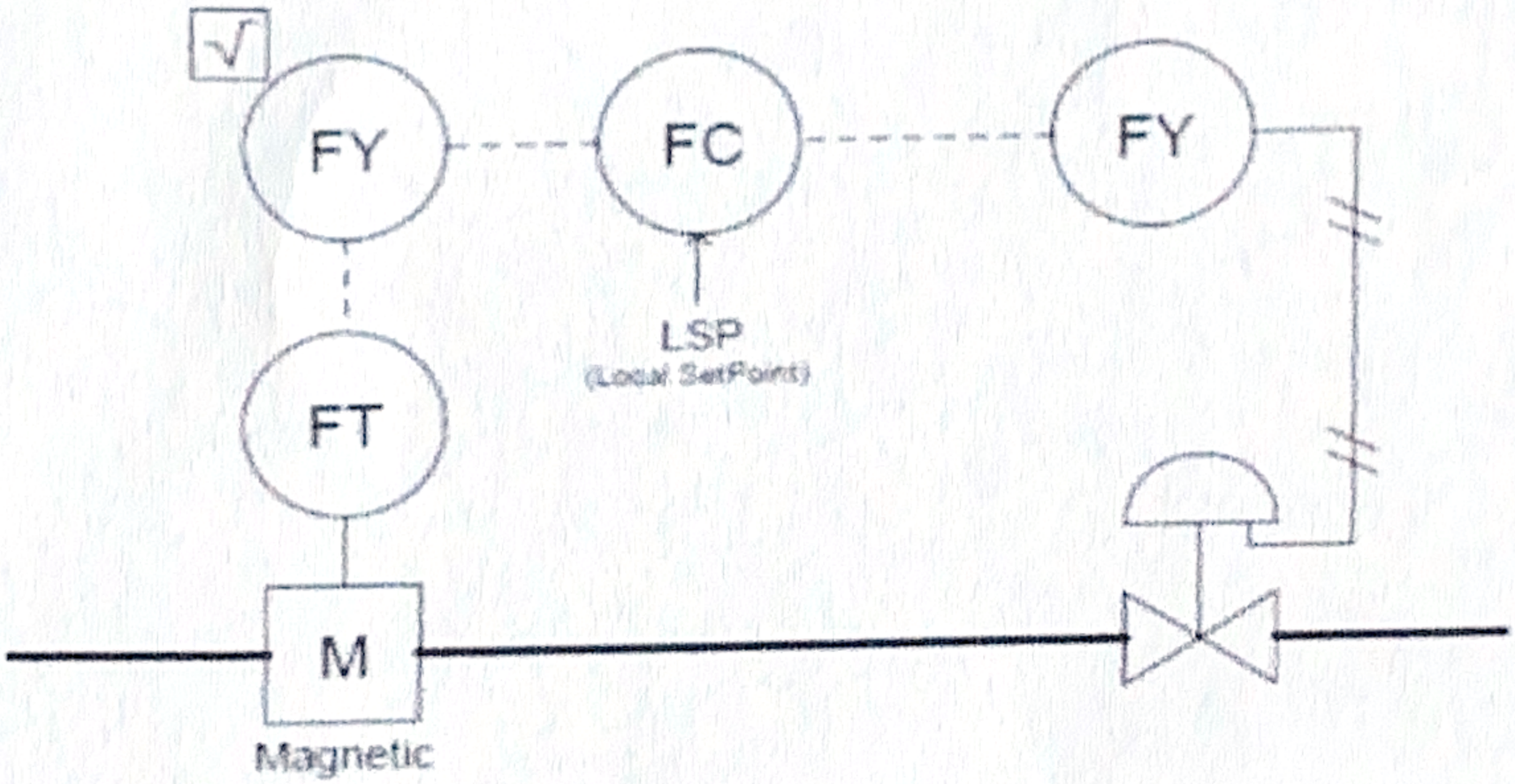
Explain what is wrong with the installation shown in FIGURE 1(above) and explain what can be done to fix it. (5)

**[24]**

## QUESTION 1: FLOW MEASUREMENTS

August 2022

1.1 The flow control system in FIGURE 1 refuses to maintain process flow at a steady set point. It seems ' sluggish' to respond to changes at high flow rates, and control at low flow rates is very erratic (rapid cycling in the measurement flow). Use the control scheme shown in FIGURE 1 to determine the problem.




(5)

**FIGURE 1**

1.2 The meter in QUESTION 1.1 only functions when measuring the flow of electrically conductive liquids.

Name common fluids that cannot be detected by the meter. (4)

1.3 State THREE precautions when an Annubar is used. (3)

1.4 The flow rate of a liquid through a 'V'-notch weir must be measured. The angle of the weir is 70° and the height of the liquid is 50 cm above the crest of the 'V'-notch (g = 10).

Calculate the flow rate of the liquid. (5)

1.5 Make a neat labelled sketch of a turbine flow meter. (5)

1.6 Would one need to compensate for the square root function when measuring flow with the meter in QUESTION 1.5? Substantiate your answer. (2)

**[24]**

DECEMBER 2022

## SECTION A

## QUESTION 1

1.1 Define or explain the following:

1.1.1 Turbulent flow

1.1.2 Specific gravity

1.1.3 Viscosity

1.1.4 Acidic solution

1.1.5 Inherent control

(5 × 2) (10)

1.2 Give ONE word or a term for each of the following descriptions. Write only the word or term next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 It is an irregular movement of fluid made up of many smaller currents swirling and moving in all directions.

1.2.2 It is defined as the action of the controller where the magnitude (size) of the output signal is equal to the (size) of the measured deviation.

1.2.3 An instrument used to measure the specific gravity of liquids.

1.2.4 It is defined as the measure of a fluid internal resistance to shear force.

1.2.5 An apparatus that measures humidity.

(5 × 2) (10)

1.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Inferential flowmeters are placed directly into the flow and measures the rate of flow.

1.3.2 Ultrasonic flowmeters operate on the principle of measuring the velocity of sound as it passes through the fluid flowing in a pipe.

1.3.3 Derivative action time is defined as the time it takes for the proportional action to repeat the derivative action.

1.3.4 Expansibility factor for flow has the value of more than one for incompressible fluids.

## SECTION A

APRIL 2023

## QUESTION 1

1.1 Explain each of the following terms:

1.1.1 Potential energy

1.1.2 Relative humidity

1.1.3 Gain

1.1.4 Polarisation

1.1.5 Stoke's law

(5 × 2) (10)

1.2 Give ONE term for each of the following descriptions by writing it next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Point where flow tempo is at its maximum and pressure at its minimum

1.2.2 Circular metal plate with a central hole with the circumference of the plate

1.2.3 Vertical tube with a conical shape, the area gradually expanding from the bottom to the top and in it a disc free to move

1.2.4 Temperature at which saturation vapour pressure equals partial pressure of water vapour in the atmosphere

1.2.5 When fluids are deformed by strain, the ratio between shear rate and shear stress will not be a constant value.

(5 × 2) (10)

QUESTION 1

1.1 Define the following terms:

1.1.1 Offset

1.1.2 Vena contracta

1.1.3 Absorption

1.1.4 Poise

1.1.5 Density

(5 × 2) (10)

1.2 Give ONE term for each of the following descriptions by writing it next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 The temperature of fluid when condensation takes place

1.2.2 Operates on the principle that when a conductor cuts a magnetic a field, an emf is induced in the conductor

1.2.3 When these fluids are deformed by strain, the ratio between shear rate and shear stress will be a constant value

1.2.4 Tapping situated 25,4 mm from the upstream and 25,4 mm from the downstream face of the orifice plate

1.2.5 A control loop where a correction is made after an error has been detected

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 An electromagnetic flow meter is capable of measuring the volumetric flow rate of non-conductivity liquids.

1.3.2 The Reynolds number is a dimensionless criterion that expresses the ratio of inertial forces to viscosity forces in a fluid flowing in a closed system.

1.3.3 An open-loop control system does not need a human operator to take corrective measures.

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1.3.4 The relationship between the pipe diameter and the diameter of the hole in the orifice plate cannot influence the calibration of a gauge.

1.3.5 When the number of hydrogen ions exceeds the number of hydroxyl ions, the solution is acidic.

(5 × 1) (5)
[20]

## SECTION A: FLOW MEASUREMENTS

Nov 2023

### QUESTION 1

1.1 Nutating-disc meters are used extensively for residential water service measurement, but can be used in many industrial applications.

1.1.1 Make a neat labelled sketch of nutating-disc flow meter. (4)

1.1.2 State THREE advantages and THREE disadvantages of nutating-disc flow meters. (6)

1.2 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (1.2.1–1.2.6) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 1.2.1 | This system will generate a signal to the process flow control valve to rectify the fault | A | primary instrument |
| 1.2.2 | This will measure the flow rate over a certain period | B | square root problem |
| 1.2.3 | This instrument generates a differential pressure of the flow | C | process control |
| 1.2.4 | This valve controls the amount of flow of a substance | D | process flow control valve |
| 1.2.5 | This controls the process for an output for an instrument and if an error occurs, to the feedback | E | integration |
| 1.2.6 | The flow rate is the square root of the differential pressure | F | feedback |

(6 × 1) (6)

**[16]**

### QUESTION 2: FLOW MEASUREMENTS

2.1 Sketch and describe the working of a drag-body flow meter. (6)

2.2 State FIVE advantages and TWO disadvantages of a drag-body flow meter. (7)

**[13]**

## SECTION A: FLOW MEASUREMENT

APRIL 2024

### QUESTION 1

1.1 Name the TWO methods that must be incorporated into secondary devices to compensate for the square root problem. (2)

1.2 Name THREE types of integrators. (3)

1.3 Make a neat, labelled sketch of an ultrasonic flow meter. (5)

1.4 State THREE advantages and TWO disadvantages of an orifice plate. (5)
**[15]**

### QUESTION 2

2.1 Explain what is meant by the *square root problem*, with reference to differential pressure and flow. (4)

2.2 Explain the working principle of a variable area flowmeter. (6)

2.3 Where do we take the reading with the meter mentioned in QUESTION 2.2? (2)

2.4 What is an integral orifice differential pressure cell, and where can it be used? (3)

2.5 Choose a term from COLUMN B that description in COLUMN A. Write only the letter (A–F) next to the question number (2.5.1–2.5.6) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 2.5.1 | The purpose of this instrument is to generate a differential of the flow. | A | feedback loop |
| 2.5.2 | The instrument that measures the total volume of flow in a certain time. | B | square root problem |
| 2.5.3 | The system will send a signal to the process flow control valve to be corrected. | C | primary instrument |
| 2.5.4 | This controls the process for an output for an instrument and if an error occurs to the feedback system. | D | control valve |
| 2.5.5 | The flow rate is the square root of the differential pressure. | E | integrator |
| 2.5.6 | The instrument that controls the amount of flow of substance. | F | process control |

(6 × 1) (6)
**[21]**

## SECTION A: FLOW MEASUREMENT

AUGUST 2024

### QUESTION 1

1.1 Name THREE types of Pitot tubes. (3)

1.2 Name FOUR precautionary measures to be considered when using a Pitot tube (4)

1.3 Define each of the following terms:

1.3.1 Turbulent flow (2)

1.3.2 Laminar flow (2)

1.4 The Venturi tube is a highly accurate primary measuring device used to create a difference in pressure in flowing fluid.

1.4.1 Make a neat, labelled sketch of a Venturi tube. (5)

1.4.2 What is the main factor limiting the size of the instrument in QUESTION 1.4.1? (1)

**[17]**

### QUESTION 2

2.1 State FOUR precautions to observe during the installation of weirs. (4)

2.2 Give ONE word or term for each of the following descriptions. Write only the answer next to the question number (2.2.1–2.2.6) in the ANSWER BOOK

2.2.1 The device that controls the amount of flow of a substance.

2.2.2 The position where the pressure reaches a minimum value after the liquid has passed through the restriction.

2.2.3 An obstruction placed in an open channel at a right angle to the direction of flow to create a pressure head.

2.2.4 An instrument that measures the total volume of flow in a certain time.

2.2.5 A velocity-sensitive device that measures the volumetric flow of gases and liquids.

2.2.6 The device that creates a difference in pressure in a flowing fluid.

(6 × 1) (6)

2.3 Name TWO types of weirs. (2)

**[12]**

DECEMBER 2024

## SECTION A: FLOW MEASUREMENT

### QUESTION 1

1.1 In the variable area flowmeter, the differential pressure across a variable aperture is used to adjust the area of the aperture.

1.1.1 Name TWO typical types of variable area meter. (2)

1.1.2 Float shapes depend on the type of fluid to be measured.

Name THREE types of float used in variable area flow meters. (3)

1.2 Electromagnetic flowmeters, sometimes known just as magnetic flowmeters, are limited to measuring the volume flow rate of electrically conductive fluids.

1.2.1 State Faraday's law of electromagnetic induction. (3)

1.2.2 Make a neat labelled sketch of magnetic flow meter. (7)

1.2.3 Write down a mathematical expression for the induced voltage of a magnetic flow meter. ALL the symbols used must be define. (5)

**[20]**

### QUESTION 2

2.1 What is the function of an integrator? (2)

2.2 Can the magnetic flowmeter measure the flow of fluids like steam, air or gases? (2)

2.3 Make a graphical sketch showing how pressure drops across an orifice plate. (5)

2.4 Define the following pressure tapping's with reference to their positions from the orifice plate.

2.4.1 Radius taps

2.4.2 Vena contractor taps

2.4.3 Corner taps

(3 × 2) (6)

**[15]**

4/1

**QUESTION 1**

1.1 Explain the difference between the rectangular, Cippoletti and V-notch open channel flowmeter. (3)

1.2 Name TWO main categories of open channel flow flowmeters. (2)

1.3 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.3.1–1.3.4) in the ANSWER BOOK.

1.3.1 Which of the following flow-measuring elements is inherently linear and requires no signal characterisation (e.g. square-root extraction) anywhere in the loop?

A Target flowmeter
B Venturi tube
C Orifice plate
D Turbine flowmeter

1.3.2 Bernoulli's Equation is a mathematical expression of:

A The ratio of kinetic to viscous forces in a flow stream
B Friction loss as fluid moves through a rough pipe
C Potential and kinetic energies in a flow stream
D Vertical height and pressure for a static fluid

1.3.3 As an incompressible fluid moves through a restriction,

A velocity increases and pressure decreases.
B velocity decreases and pressure increases.
C velocity decreases and pressure remains the same.
D both velocity and pressure increase.

1.3.4 What type of dynamic fluid effect is illustrated by a flag flapping in the breeze?

A Transitional flow
B Laminar flow
C Vortex shedding
D Cavitation

(4 × 2) (8)

1.4 How do you identify an orifice in the pipeline? (2)

1.5 Why does flow measurement in Bernoulli primary flow device have to be rooted? (3)

**[18]**

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**QUESTION 2**

Calculate the bore inside diameter of an orifice plate stainless steel pipe.

- D + D/2 taps
- Maximum flow of gas is 740 $m^3/h$
- Diameter of the pipe is 250 mm
- Maximum differential pressure is 1 000 Pa
- Density of the gas is 6,408 $kg/m^3$
- The pipeline pressure is controlled at 375 mm W.G (3,6764 Pa)
- Gas temperature is 30 °C
- The specific heat ratio is 1,4
- Area relation is 0,32

**[12]**

**TOTAL SECTION A: 30**

## SECTION A: FLOW MEASUREMENT

AUGUST 2025

### QUESTION 1

1.1 The flow rate of fluid flow can be expressed in two ways, namely volume flow rate and mass flow rate.

1.1.1 Explain the difference between *volume flow rate* and *mass flow rate*.

1.1.2 Write down the mathematical formulae for volume flow rate and for mass flow rate.

(2 × 2) (4)

1.2 There are two major types of ultrasonic flowmeter measuring principles in use today, the transit-time method and the Doppler shift method.

1.2.1 Briefly describe the construction of an ultrasonic flowmeter. (4)

1.2.2 Explain the principle for the transit-time ultrasonic flowmeter. (6)

**[14]**

### QUESTION 2

Calculate the flow rate of water in a pipeline, based on the following information:

Maximum differential pressure = 2 000 Pa
Inside diameter of orifice plate = 20 cm
Outside diameter of orifice plate = 30 cm
Line pressure = 30 kPa
Water temperature = 30 °C
D + D/2 tapping are used.

2.1 Calculate the area relation (m). (2)

2.2 Determine the discharge coefficient. (1)

2.3 Calculate the velocity approach factor (E). (2)

2.4 What is the value of the expansion factor (ε)? Give a reason for your answer. (2)

2.5 Calculate Reynold s number ($R_d$). (5)

2.6 Calculate the correction factor (Z). (3)

2.7 Calculate the flow rate of water. (3)

**[18]**

**TOTAL SECTION A: 32**

## SECTION A DECEMBER 2025

### QUESTION 1: FLOW MEASUREMENT

1.1 The orifice meter is a common and simple method to measure flowrate. It consists of a thin, steel plate with a hole through which the flow passes.

1.1.1 Describe the construction of a segmental orifice plate. (2)

1.1.2 Explain why this orifice is used where dirty fluids are encountered. (1)

1.1.3 Name THREE other differential pressure primary devices. (3)

1.2 What is a *weir*? (2)

1.3 Instruments used in the measurement of flow can be divided into two main classes; namely rate of flow meters and volumetric meters:

1.3.1 Explain the operational differences between these classes of meters. (2 × 2) (4)

1.3.2 Each instrument may be considered as two parts, the primary element and the secondary element. Define the two elements. (2)

1.3.3 What is the purpose of measuring flow elements? (1)

**[15]**

### QUESTION 2

There are basically two types of thermal mass flow meters: those that measure temperature rise of fluid, and those that measure the rate of heat loss to the fluid stream.

2.1 Make a neat, labelled sketch of a thermal mass flow meter that measures temperature rise of a fluid. (5)

2.2 Explain the working principle of this heat transfer flow meter. (7)

2.3 What is the disadvantage of this flow meter? (2)

2.4 What is the advantage of the rate of heat loss flow meter over the heat transfer flow meter? (1)

**[15]**

**TOTAL SECTION A: 30**

AUGUST 2026

1.3 The following information is known about the TWO weirs in FIGURE 2 below. The width of both weirs is 3 m, and the height is 4 m.

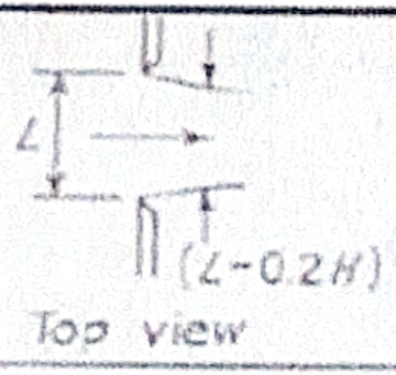
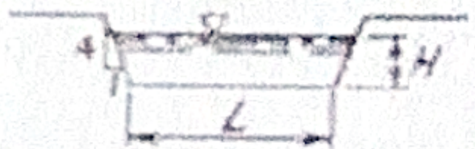
| | | | |
|---|---|---|---|
| Rectangular Weir (with contraction) |   Top view | Side view | $Q = 3.33(L - 0.2H)H^{3/2}$ |
| Trapezoidal Weir |   End view | | $Q = 3.37LH^{3/2}$ |

**FIGURE 2**

1.3.1 Calculate the flow rate of the *rectangular* weir. (2)

1.3.2 Calculate the flow rate of the *trapezoidal* weir. (2)

1.3.3 Comment on the answers. (1)

**[20]**

**SECTION A:** AUGUST 2026

## QUESTION 1: FLOW MEASUREMENT

1.1 Define the following terms:

1.1.1 Laminar flow

1.1.2 Turbulent flow

1.1.3 Incompressible fluid

1.1.4 Compressible fluid

(4 × 2) (8)

1.2 Study the diagram in FIGURE 1 below, and answer the following questions:

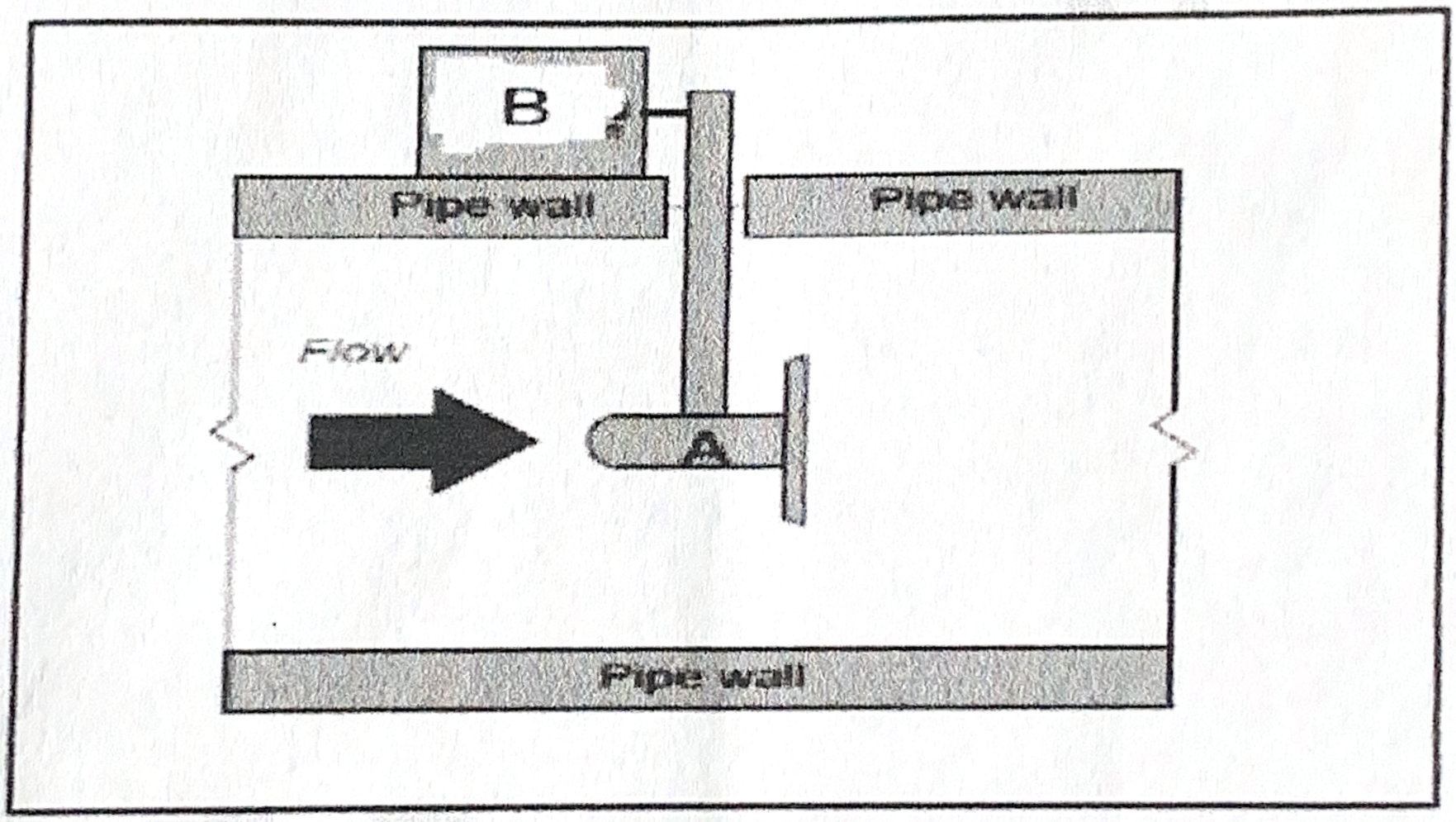



**FIGURE 1**

1.2.1 Name the flow meter shown above. (1)

1.2.2 Name the parts marked A and B. (2)

1.2.3 Explain the working principle of this flow meter. (4)